

NAME: _____

PERIOD: _____

Two Library Survey Plans

AP Statistics · Unit 3 · Topic 3.3

[STAR] THE PROBLEM

A library system has 20,000 adult cardholders and wants to estimate p , the proportion who used a digital audiobook last month.

Plan A: A computer takes an SRS of 200 cardholder IDs. All answer; 116 say yes.

Plan B: A newsletter link lets cardholders choose to respond once. Of 800 respondents, 520 say yes.

Eli: “Use Plan B. Its larger sample gives the narrower interval.”

Dana: “Use Plan A. Entry into the sample matters more than the narrowest output.”

The 95% formula output for B is (0.6169, 0.6831), width 0.0661. A’s formula width is 0.1368.

Plan counts

Plan	N	n	Yes	No
A	20,000	200	116	84
B	20,000	800	520	280

FIRST TAKE — one sentence before you work the parts

Which plan, if either, should estimate all adult cardholders? Cite one collection detail.

[BOOK] ONE-SAMPLE INTERVAL FOR A PROPORTION (keep on the page)

Use a one-sample z -interval for p : $\hat{p} \pm z^* \sqrt{\hat{p}(1 - \hat{p})/n}$; for 95% confidence, $z^* = 1.96$. Check (1) random data collection, (2) when sampling without replacement, $n \leq 0.10N$, and (3) at least 10 observed successes and 10 failures. A smaller formula margin of error describes less random sampling variation; it does not repair systematic error caused by who chooses to respond.

Work (a)–(c).

DOK 1 (a) Define the population parameter p in context and name the confidence-interval procedure that would be appropriate when its conditions are met.

DOK 2 (b) For Plan A, calculate $\hat{p}$ and its 95% interval. Check A's random, 10 percent, and success/failure conditions. For Plan B, identify the condition that fails.

DOK 3 (c) In 3–4 sentences, choose the plan that can estimate all cardholders. Use recruitment and one numerical condition. Explain why the supplied widths do or do not support Eli's recommendation.

Frame: I would report Plan ____ because _____, and the numerical check _____ is satisfied.

Frame: Although Plan B's formula width is _____, it would mislead readers because _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.